Day 5 morning — Regression as a generative model

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Monday 16 November 2026 · Pontoppidanstræde 101, room 1.001

A mean that depends on a predictor · scales and centring · priors you can picture · prior predictive simulation · regression in Stan

Today

08:45	**Lecture A**	Regression as a generative model · scales · priors · prior predictive
09:45	**Exercise A**	Your own timing data: scales, priors, prior predictive · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Fit · mean vs new attempt · posterior predictive checks · robust
13:30	**Exercise B**	Fit, predict, check, revise the model · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

Follow-up from yesterday

From the Day 4 diaries and plenary

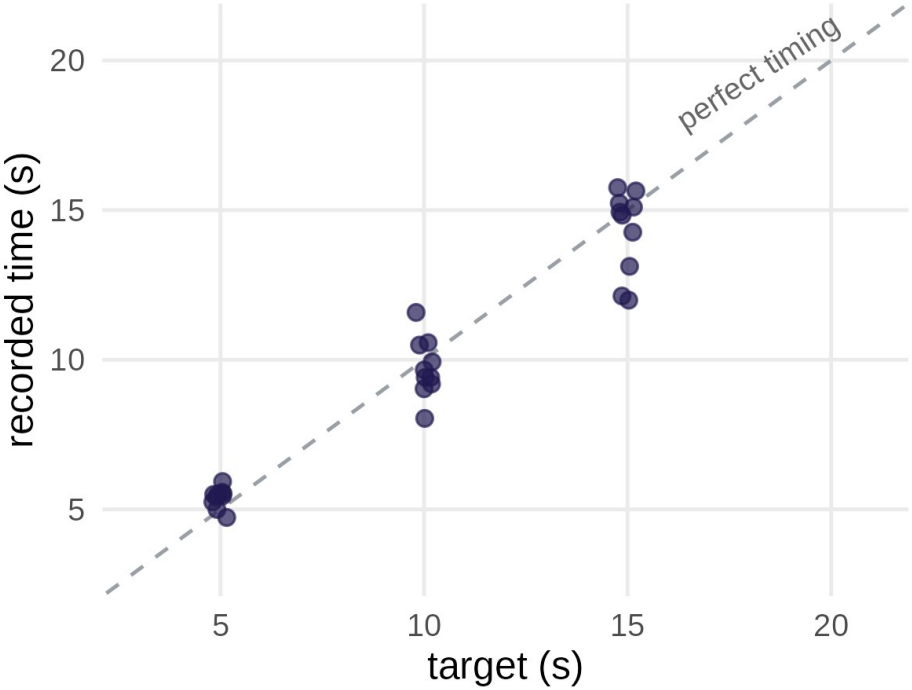
Fill in during the weekend: the two or three points that came up most often under “not yet familiar”.

1. …
2. …
3. …

By the end of this lecture you can

1. **Write** a linear regression as a model that generates data
2. **Choose** a scale for the predictor so that the parameters mean something
3. **Set** priors on that scale and check them by prior predictive simulation
4. **Write** the regression in Stan
5. **Interpret** intercept, slope and spread in the units of the data

Does her clock scale?



One person's 30 attempts from the Day 1 time game: ten each at 5, 10 and 15 s, no feedback (illustrative numbers).

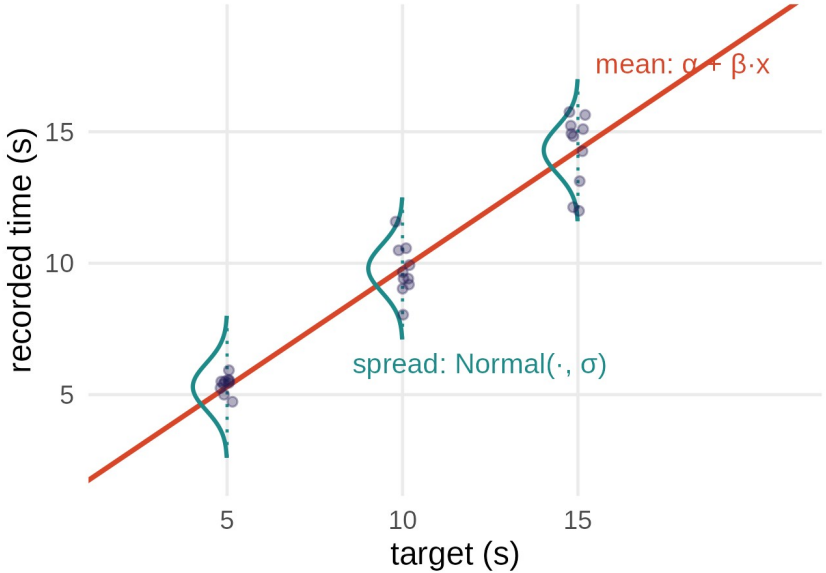
Questions

1. How does her recorded time grow with the target?
2. Does she fall short at long targets?
3. What would she record at 20 s, which she never tried?

One quantity depends on another: a regression.

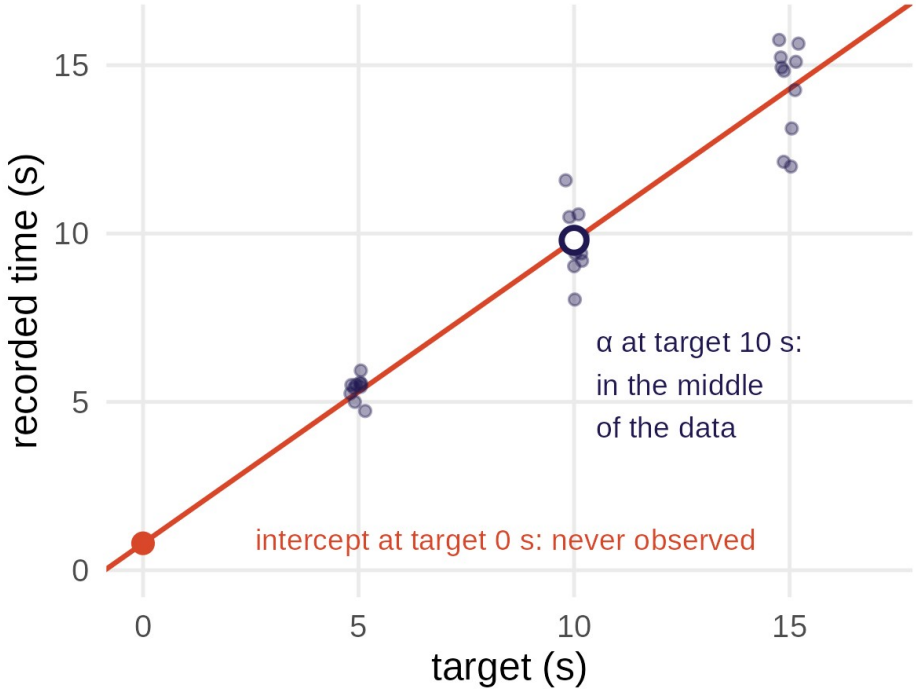
Regression: a model that generates the data

$$y_i \sim \text{Normal}(\alpha + \beta \cdot x_i, \sigma), \quad \text{independent given } x$$



- The mean of y depends on x through a straight line: $\alpha + \beta \cdot x$.
- Around that mean, attempts scatter with SD σ — the same at every x.
- To simulate data: pick x, compute the mean, draw y from the normal.
- Day 4's normal model is the special case $\beta = 0$.

Choose a scale for x



$x_i = target_i - 10$

- α: her average recorded time at a 10-s target.
- β: extra recorded seconds per extra target second. β = 1: perfect scaling; β < 1: long targets fall short.
- With x = target, α would be her time at a 0-s target: never observed, hard to set a prior for.

Centre the predictor in the middle of the data: every parameter then has a meaning.

Priors on that scale

parameter	prior	what it says
α (s)	Normal(10, 2)	at a 10-s target she records about 10 s, probably 6–14
β	Normal(1, 0.5)	recorded time grows with the target, roughly one-to-one; 0 to 2 is possible
σ (s)	Normal^{+}(0, 2)	attempts scatter by less than about 4 s

Each prior is about something we can picture: a time in seconds, a ratio of times, a spread in seconds.

Does this prior make sense? Simulate data from it and look.

Prior predictive simulation

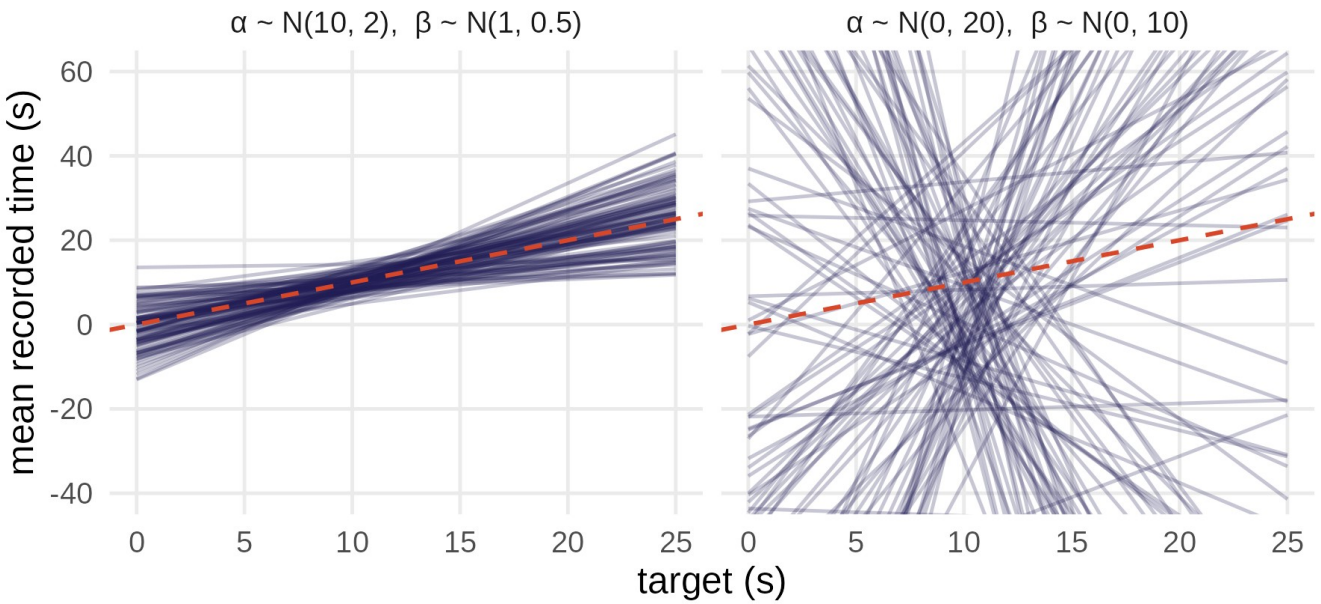
1. Draw α, β and σ from their priors.
2. Compute the mean $\alpha + \beta \cdot x$ at each target.
3. Draw attempts y around the mean with SD σ.
4. Look: are these plausible recorded times?

```
S <- 4000
alpha <- rnorm(S, 10, 2)
beta  <- rnorm(S, 1, 0.5)
sigma <- abs(rnorm(S, 0, 2))

x <- 15 - 10     # target 15 s, centred
y_sim <- rnorm(S, alpha + beta * x, sigma)
quantile(y_sim, c(0.05, 0.95))
```

No data are used: we see what the model expects before it has seen anything.

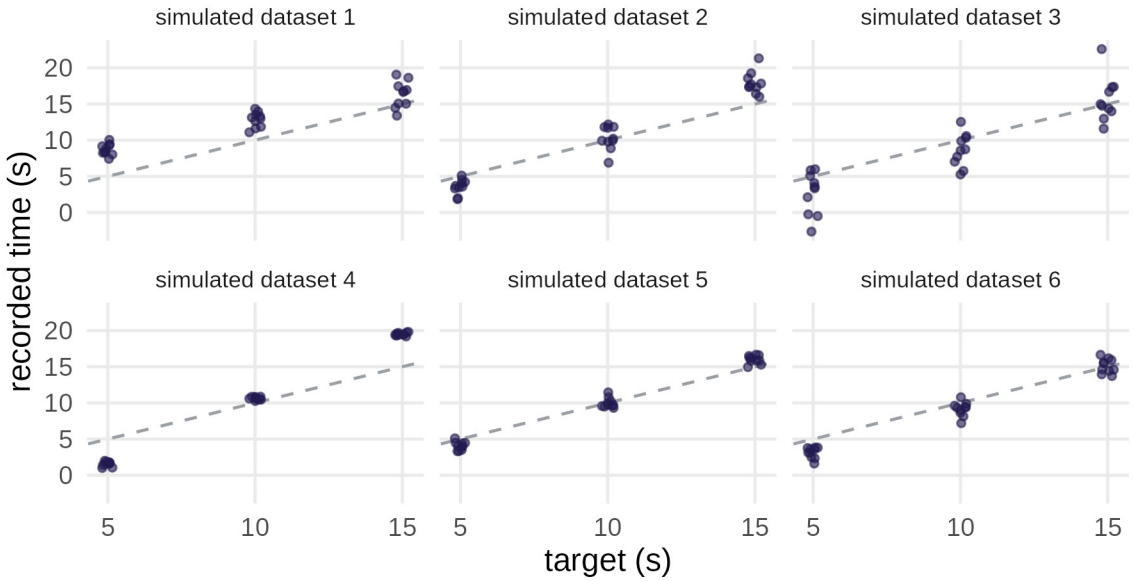
What the prior predicts



Each line: the mean $\alpha + \beta \cdot x$ for one prior draw. Dashed red: perfect timing.

Left: lines that look like timing. Right: a "vague" prior expects impossible recorded times.

Fake data from the prior



target	90% of prior attempts	below 0 s
5 s	−1.1 – 11.1	9%
10 s	5.5 – 14.6	0.2%
15 s	8.8 – 21.3	0%
20 s	10.5 – 29.5	0%

- Mostly plausible attempts.
- At 5 s, 9% are negative: the prior is a little too wide there. With 30 attempts the data will dominate; a tighter σ prior would fix it.

The regression in Stan

```
data {
  int<lower=1> N;
  vector[N] x;              // target - 10 (s)
  vector[N] y;              // recorded time (s)
}
parameters {
  real alpha;
  real beta;
  real<lower=0> sigma;
}
model {
  alpha ~ normal(10, 2);
  beta ~ normal(1, 0.5);
  sigma ~ normal(0, 2);
  y ~ normal(alpha + beta * x, sigma);
}
```

- alpha + beta * x is a vector: one mean per attempt.
- The likelihood line is the model on the previous slides, word for word.
- We centre x in R: x = target − 10, before passing the data.
  - This afternoon: generated quantities for predictions and checks.

Worked example: reading a regression

Question 1

Suppose $\alpha = 9.8$ and $\beta = 0.9$.

What does she record on average at 5, 10 and 15 s? Where does she fall short?

Question 2

Without centring (x = target), what would α mean? What prior would you give it?

Question 3

A colleague proposes $\beta \sim \text{Normal}(0, 10)$ “to let the data speak”. What does the prior predictive simulation show, and what is wrong with it?

Worked example: answers

Question 1

5 s: $9.8 + 0.9 \times (-5) = 5.3$

10 s: 9.8

15 s: $9.8 + 0.9 \times 5 = 14.3$

Short targets a little too long, long targets too short: her clock compresses ($\beta < 1$).

Question 2

α would be her mean recorded time at a 0-s target: outside the data, physically odd. Its prior would have to come from the slope as well: α and β become strongly linked. Centring avoids both problems.

Question 3

Half the prior lines fall with the target, and most predict recorded times far below 0 or above 40 s. The prior is not "neutral": it expects impossible data. Use what you know: times grow with the target.

Exercise A · 09:45 – 11:10

- Plot recorded time against target for one person from the time game.
- Centre the target, and write down what α, β and σ mean in seconds.
- Prior predictive simulation: lines and fake data; is the prior plausible?
- Read the Stan program you will fit this afternoon.
- Write your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

11:10

peer review A

Take a break when it suits your group. Rotate roles: coder · checker · question writers · interviewer · reporter.

Peer review A · 11:10 – 11:55

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

2 min

Wrap-up

What to take into lunch and Lecture B.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

```
sample(not_yet_drawn, 2)   # plenary draw
```